Interview Assignment: Build an AI Agent

At DataRobot, we believe the future of AI is agentic. Our strategy is centered on empowering our customers to build and deploy AI agents that solve their most complex business challenges. The Professional Services organization is at the heart of this mission, responsible for driving customer adoption and value realization with a specific focus on our agentic offerings.

As the incoming member of this team, you will serve as a key evangelist for this transformation. This assignment is designed to give you a hands-on opportunity to engage with this core part of our strategy and demonstrate your vision, creativity, and technical leadership.

The Assignment

Your task is to build a functioning AI agent and prepare a short presentation for your next interview.

The agent should be designed to solve a problem you are passionate about. This could be anything from a personal productivity tool (e.g., an agent that summarizes your unread emails and drafts responses) to a more complex business process automation (e.g., an agent that analyzes customer support tickets and suggests resource allocation).

IMPORTANT NOTE: The final output should be presented as a complete application, including both frontend and backend components.

Your Mission

1. Identify a Problem: Choose a real-world problem or inefficiency that an AI agent could meaningfully address.
2. Build Your Agent: You are free to use open-source frameworks like LangChain, LangGraph, CrewAI, Pydantic or LlamaIndex that you are most comfortable with, but please use a "code first" approach.

3. Prepare Your Presentation: Prepare a 15-minute presentation and live demo of your agent. Be ready to discuss:
    - The Problem: What business or personal challenge does your agent solve?
    - Your Solution: Briefly explain the agent's architecture, the tools you used, and why you chose them.
    - Live Demo: Showcase the agent in action.
    - Challenges & Learnings: What obstacles did you face, and what did you learn?
    - The Vision: How would you apply this thinking to solve a DataRobot customer's problem? How could a solution like this be scaled and productized?

What We're Looking For

This is not just a test of coding ability. We are most interested in your approach and strategic thinking. We will be looking for:

- Problem-Solving Acumen: The clarity and relevance of the problem you chose to solve.
- Strategic Vision: Your ability to connect a specific solution to broader business value and customer outcomes.
- Technical Credibility: A functional demo and a clear explanation of your technical choices.
- Creativity & Passion: Your enthusiasm for the potential of agentic AI and your creativity in applying it.

We understand you are busy, and we are not expecting a production-ready application. The goal is to see how you think and build. This exercise will serve as the foundation for a rich, strategic discussion in our next conversation.

Submission

Please submit your code (GitHub preferred) and presentation deck via email for the team to review. Once received, the team will review your submission and follow up with next steps.

We look forward to seeing what you create.